

Population genomics of the endangered Peruvian algarrobo (*Neltuma pallida*) reveals low genetic diversity and a recent bottleneck in a protected dry-forest population

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Abstract

The Pomac Forest Historic Sanctuary (SHBP), a protected area on the northern coast of Peru, conserves biodiversity representative of Equatorial dry forests. SHBP is mainly composed of algarrobo (*Neltuma pallida*), a species adapted to the variable climate conditions and known for its multiple ecological functions and ecosystem services. Algarrobos face an unprecedented population decline despite their remarkable natural adaptation, worsened by their ongoing habitat degradation and overexploitation. In this study, we aim to characterize the genomic diversity of the protected forest in the SHBP to establish a baseline for future monitoring and research. Results show that the algarrobo population in the SHBP is unstructured, has low inbreeding indices, an average autosomal heterozygosity of 0.19%, and a per-site nucleotide diversity of 0.0023. Patterns of low genetic diversity are associated with a recent bottleneck decline starting around four generations ago. Potentially deleterious variants with varying effects, possibly playing a role in population adaptability were predicted. In conclusion, our results provide reference genetic indices for future monitoring of algarrobo in its native range and raises questions about the impact of recent habitat degradation on its genome.

Keywords: Algarrobo, Carob, heterozygosity, inbreeding, Lambayeque, mesquite, multivariate analyses, Pomac, relatedness

Background

Seasonally dry tropical forests (STDFs) are among the most endangered biomes on the planet [1,2] due to their long history of anthropogenic transformation and degradation [3]. STDFs are distinctive and important ecosystems for Peruvian biodiversity, covering around 40,253.38 km² of territory [4]. Among their many ecological roles, STDFs are recognized for sustaining high diversity and endemism [2,5,6], harboring complex interactions, and providing a variety of ecosystem services [7].

The Peruvian algarrobo, *Neltuma pallida* (ex. *Prosopis pallida*) (Humb. & Bonpl. ex Willd.) C.E.Hughes & G.P.Lewis (Fabaceae), is the predominant species in low-elevation (0 – 700 masl) STDFs (covering 37.48% of the total STDFs extension) along the Peruvian coast [4,8]. As algarrobos thrive under the arid and variable climate conditions of coastal STDFs, their complex system of nitrogen-fixing roots and dense canopies create a fertile and moist microclimate suitable for the survival of plants and animals, including human communities [9,10]. Algarrobo-derived resources have also been used by humans for millennia, particularly wood for construction and fuel, which currently represents a major driver of overexploitation for these species [11,12]. In addition to its keystone role in the ecosystem, the Peruvian algarrobo is a long-lived species, with individuals living more than 100 years [13,14]. For example, a remarkable individual (“Algarrobo Rey”) in northern Peru, with a trunk diameter of 8.40 m, is thought to be more than 500 years old (W. Zeña, personal communication, June 2025). Meanwhile, in the south, relict trees

reach even larger sizes, with one notable example (“Huarango Milenario”) that has a trunk diameter of nearly 5 m and has been determined to be 1,064 years old [15].

In recent years, algarrobos in low-elevation STDFs have experienced an unprecedented dieback compounding existing threats. Up to 95% of algarrobo forests in northern Peru have been impacted by an unknown disease that, as of 2019, had affected more than half of the population [16,17]. The emergence of plagues such as insects, fungal, viral, and oomycete pathogens is among the biotic factors associated with the algarrobo decline [18-23]. However, the most significant threats to algarrobo are anthropogenic, with land-use change, overgrazing, and overexploitation as the primary drivers for their ongoing degradation [24]. As a keystone species of SDTFs in Peru, the historic human-led algarrobo’s decline has resulted in irreparable desertification on the southern coast [11,25]. Currently, algarrobo is almost extinct in the hyper-arid southern coast, and populations in the northern coast are rapidly declining despite El Niño-Southern Oscillation (ENSO)-driven regrowth [26-28].

STDFs degradation and algarrobo decline are expected to have an impact on the species’ genetic diversity, which reforestation efforts may be insufficient to reverse. Additionally, algarrobo populations remain uncharacterized at the molecular level, precluding precise assessments of their endangerment and conservation efforts. [29]. In this study, we aimed to understand the effects of the algarrobo decline on its genetic diversity by using a genome reference-guided population genomics approach, with whole-genome sequencing data and high sampling density. We focused on the Historic Sanctuary Pomac Forest (SHBP), a state-protected area in northern Peru holding a representative sample of the Equatorial STDF. By evaluating the genetic diversity status in the SHBP,

we establish a baseline for future monitoring and a reference for the evaluation of other less-protected algarrobo populations in equal or greater endangerment.

Methods

Study site and sample collection

Samples were collected in the SHBP, located in the lower valley of the seasonal river La Leche, within the provinces of Ferreñafe and Lambayeque (Lambayeque geographical department, northwestern Peru) (Figure 1). SHBP is a natural protected area covering 58.87 km² and dominated by algarrobo, with areas reaching densities up to 30,700 trees/km² [30]. Furthermore, *N. pallida* is the sole representative of the genus in the study area.

The arid climate in SHBP is extremely variable, with temperatures going from a minimum of 11.5 °C to a maximum of 34.4 °C, and a yearly precipitation of only 107.8 mm, mainly concentrated between March and April [31]. ENSO (El Niño-Southern Oscillation) is a natural climate cycle that happens every 2-7 years in the Pacific Ocean, changing the ocean temperatures and air pressure, resulting in torrential rains of up to 1,449.5 mm in the Lambayeque region [31].

Samples were collected in SHBP from 31 transects installed at 100-meter intervals, with 3 individuals per transect, resulting in a total of 93 samples across all sectors (Figure 1, Supplementary Table 1). Two to three compound leaves of each tree were collected, rinsed with distilled water, dried with paper towels, and stored fresh in bags containing silica gel. The material used in this study was collected under the following permits: RJ N° 003-2020-SERNANP-JEF and RJ N°002-2022-SERNANP-JEF. No live material was

exported out of the country. All plant material was initially identified in the field by Renato La Torre, Manuel Saucedo-Bazalar, Gisella Orjeda, and expert local park rangers. These identifications were subsequently verified in a laboratory setting by Renato La Torre and Esteban Caycho. A representative voucher specimen of *Neltuma pallida* was deposited in the Herbario San Marcos (USM) of the Museo de Historia Natural – UNMSM in Lima, Peru, under the accession number USM 335439.

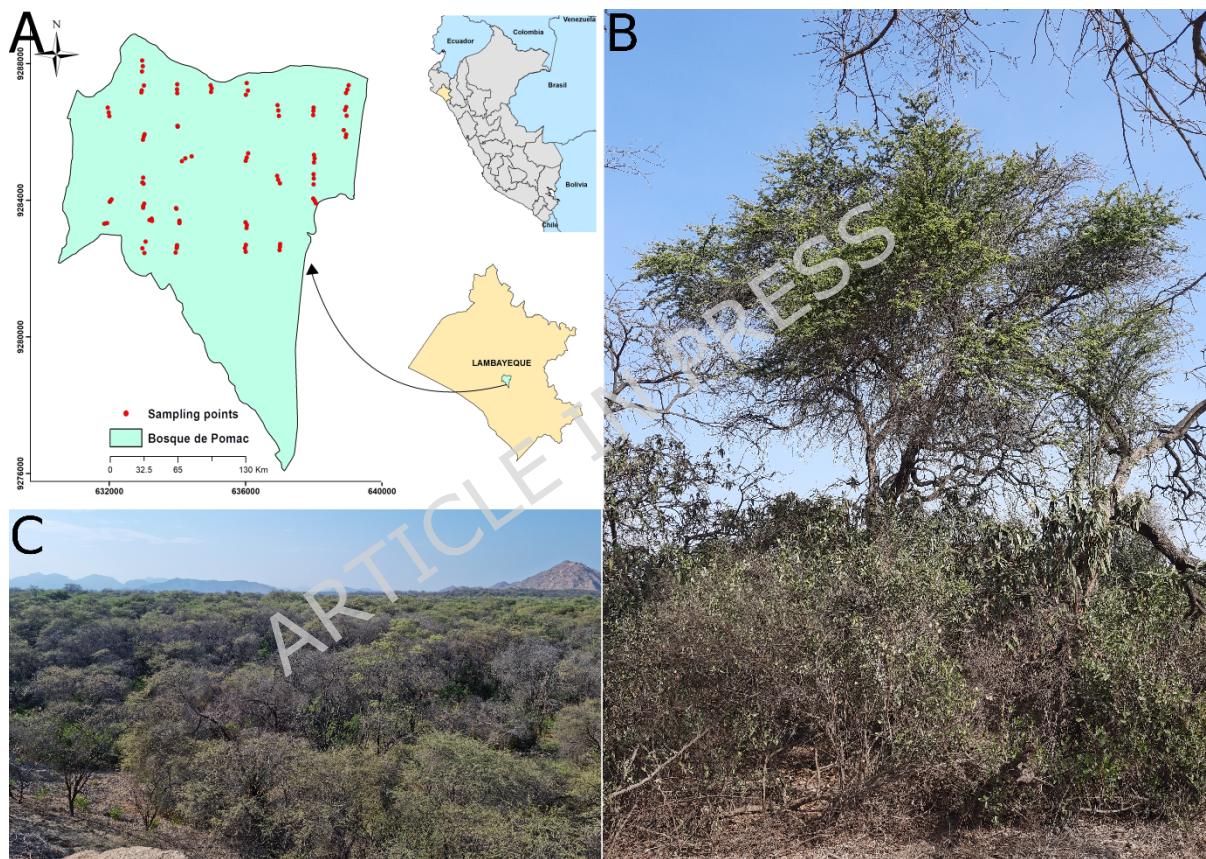


Figure 1. Distribution of samples collected across the SHBP. A. A total of 93 samples was collected from the different zones in the protected area Historic Sanctuary Pomac Forest (SHBP). **B.** One of the adult individuals collected for this study. **C.** Many of the trees in the SHBP dry forest are diseased or already dead.

DNA extraction and whole-genome sequencing

All samples were ground with liquid nitrogen using individual sterile mortars and pestles, and stored at -80 °C until processing. For DNA extraction, up to 100 mg of ground material was processed using a CTAB protocol [32]. An additional purification step was performed by adding an appropriate volume of 5 M NaCl to reach a final concentration of 2 M NaCl, followed by the addition of 1× volume of absolute ethanol. The mixture was centrifuged at 12,000 rpm for 5 minutes, the resulting pellet was dried for 15 minutes, and finally solubilized in 50 µl of nuclease-free water. This purification step was particularly useful for polysaccharide-rich samples which turned colored and viscous otherwise. Successful samples as assessed by 2% agarose gels and spectrophotometer/fluorometer quantifications were stored at -20 °C until shipped for sequencing. Whole-genome sequencing was performed through the Texas A&M AgriLife Genomics and Bioinformatics Service. Illumina libraries were sequenced on a NovaSeq 6000 in 150-bp paired-end mode.

More than 2.7 billion reads, corresponding to 810 billion bases, were generated for 93 Peruvian algarrobo samples in SHBP, with an average of 29.3 million reads and 8.8 billion bases for each sample (Supplementary Table 2). Sequencing depths ranged from 1.5X to 16.1X (mean 10.9X, SD 4.5X) for around 78.4% of the nuclear reference genome. For the chloroplast genome, sequencing depths ranged from 185.8X to 1828.2X (mean 874X, SD 442.6X) for around 69.7% of the chloroplast sequence.

Reads processing and mapping

Sequencing reads were processed and mapped to the chloroplast (NC_084206.1) and nuclear (GCA_046055785.1) *N. pallida* reference genomes [33,34] using the Paleomix v1.3.7 pipeline [35]. This pipeline removes adapters and low-quality reads using AdapterRemoval v2.3.3, merging those reads with a minimum of 11-bp overlap and retaining only reads with a minimum length of 25 bp [36]. Reads were mapped to the reference genome using bwa v0.7.17-r1188 with the mem algorithm [37]. Mapped reads with mapping quality (MAPQ) below 30 were filtered out and duplicates were removed using SAMtools v1.19.2 [38] and Picard v2.27.5 (<http://broadinstitute.github.io/picard/>), respectively.

Chloroplast genome diversity

Chloroplast haplotypes were called using GATK's HaplotypeCaller [39]. Variant calling was performed using GenotypeGVCFs and then filtered conservatively using VariantFiltration and VCFtools v0.1.17 [40]. These filters discarded variants beyond a minimum variant quality ($QUAL < 30$), minimum quality by depth ($QD < 30$), maximum symmetric odds ratio to detect strand bias ($SOR > 3$), maximum Phred-scaled p -value for exact test of excess heterozygosity ($FS > 60$), minimum root mean square of mapping qualities ($MQ < 40$), minimum Z-score from Wilcoxon rank sum test of alternative vs. reference read mapping qualities ($MQRankSum < -12.5$), and minimum Z-score from Wilcoxon rank sum test of alternative vs. reference read position bias ($ReadPosRankSum < -8$). Individuals with less than 50% of sites after filtering were removed from the analysis, leaving a total of 2,194 called and filtered variants. Consensus sequences against the reference sequence were generated for each individual from their respective

filtered vcf file using BCFtools v1.21 [41] and used for constructing a median-joining haplotype network [42] with $\epsilon = 0$ using PopART v1.7 [43].

Genotype likelihood estimation and variant filtering

Genotype likelihoods were estimated for the nuclear genome using ANGSD v0.940 [44] with the GATK method (-GL 2), setting a minimum base quality score of 20 (-minQ 20), minimum mapping quality of 30 (-minMapQ 30), removing non-primary, failure, or duplicated reads and reads with multiple best hits (-remove_bads 1, -uniqueOnly 1), downgrading mapping quality for excessive mismatches (-C 50), and performing baq computation (-baq 1). Individuals and sites with a sequencing depth less than 2 were discarded (-setMinDepthInd 2 and -geno_minDepth 2, respectively), as well as sites with data available for less than half the individuals (-minInd). Variant allele frequencies were estimated from genotype likelihoods (-doMajorMinor 1, -doMaf 3) and filtered using a $1e-6$ cutoff p -value (-SNP_pval $1e-6$) and a minimum allele frequency of 0.01 (-minMaf 0.01). Lastly, genotypes were filtered by estimating the posterior probability based on the allele frequency as a prior (-doPost 1) and setting a cutoff of 0.95 (-postCutoff 0.95).

Linkage disequilibrium (LD) was estimated across the genome using PLINK v1.90 [45] with a window size of 50 and a step of 3 variants, and sites with a pairwise LD > 0.5 were removed. Sites within repetitive regions in the reference genome annotation file were filtered out to minimize mapping errors due to complex regions. In addition, CallableLoci from GATK v3.7-0 [39] was used to identify and remove sites within regions with low or excessive sequencing depth (a third or two times the average depth, respectively) using a representative deeply-sequenced (10.93X) individual (Alg10). . Around 47.5% of the

genome was filtered out from the analysis due to being within low or excessive depth regions or repetitive regions, leaving a total of 187,629,614 sites (Table 1). Genotype likelihood-based analyses in these filtered regions resulted in 2,063,009 nuclear variant sites (MAF > 0.01). A total of 1,222,850 SNPs in strong LD ($r^2 > 0.5$) were removed for the population structure analyses, leaving 840,159 for the population structure analysis (Table 1).

Table 1. Number of nuclear sites and variants throughout the analysis.

Criteria	Number of sites	Percentage of the genome (%)
<i>N. pallida</i> genome before filtering	394,713,541	100
Ambiguous regions (Ns)	72,600	0.018
Repetitive regions	206,613,734	52.35
Too low depth regions (< 3X)	28,856,933	7.31
Excessive depth regions (> 24X)	6,388,083	1.62
Variants before region filter	6,490,562	1.64
<i>N. pallida</i> genome after region filter	187,629,614	47.54
Variants after region filter	2,063,009	0.52
Variants after pruning strongly linked sites	840,159	0.21

Relatedness, inbreeding and population structure analysis

Pairwise relatedness and inbreeding coefficients were estimated using NgsRelate v2.0 [46,47]. First- and second-degree relatives based on the coefficient of kinship ($\theta > 0.125$) were identified and one from the pair was removed from the analyses (Supplementary Table 1). A principal component analysis was carried out based on the genotype likelihoods for the unlinked sites using PCAngsd v1.35 [48]. Similarly, an estimation of

individual admixture proportions using genotype likelihoods was performed using NGSadmix v32 [49]. Ten runs were performed for each prior number of ancestral populations (K) tested, from 2 to 6, and the run with the highest likelihood was selected based on their likelihood values.

Estimation of genetic diversity indices

Site allele frequency likelihoods based on individual genotype likelihoods (GATK method, -GL 2) assuming Hardy-Weinberg Equilibrium (-doSaf 1) were calculated for each unrelated individual and for all samples using ANGSD v0.940 [44] with settings for filtering low-quality mappings as described above, but keeping all sites. The folded site frequency spectrum (SFS) was estimated using the maximum likelihood method specifying callable regions (see above) and outside repetitive elements. SFS estimations for each individual sample was used to calculate the per-sample genome-wide autosomal heterozygosity, while the multi-sample SFS was used for estimating the per-site Watterson estimator, nucleotide diversity (π), and Tajima's D using a window size of 100 kb and a step size of 10 kb using realSFS from ANGSD v0.940 [50]. Per-site genome-wide nucleotide diversity (π) was used as a proxy for expected heterozygosity (H_e), as both metrics are closely related measures of genetic diversity and converge under standard assumptions for biallelic loci. Lastly, the effect of variants on coding sequences and its likely impact was estimated using SnpEff [51]. Results were illustrated using circlize v0.4.17 in R [52].

Demography inference

Historical estimates of the effective population size were obtained using the software GONE [53]. ANGSD beagle files were filtered to keep only individuals with a minimum

coverage of 10.8 X ($n = 65$), discard sites with a recalculated MAF < 0.01 , and reformatted to be used in GONE. Input parameters included Haldane correction for genetic distances (DIST = 1), 2,000 generations (NGEN = 2000), 400 bins, and a maximum of 50,000 SNPs per chromosome to be analyzed in each run. A hundred runs were performed with 40 GONE internal replicates.

Results

Sequencing and genotyping

Inbreeding, relatedness, and population structure

Nine samples were removed due to having a low mapping success for both nuclear and chloroplast genomes, and one more for being in a closely related pair. There is no strong signal of inbreeding, with most individuals having an inbreeding coefficient (F) close to zero, and only two with F greater than 0.1 (Figure 2A). Most of the pairs of individuals have kinship coefficients below the second order ($\theta = 2^{-3} = 0.125$), even within transects only 100 meters apart. An exception included two individuals distanced 630.2 meters within the highly dense southwestern area (Figure 2B). This pair included one of the inbred individuals (Alg82, $F = 0.18$), and the relatedness signal was also supported with the KING coefficient (Supplementary Figure 1).

No genetic structure was recovered for the algarrobo population in SHBP (Figure 2C). Two individuals from the southernmost area “Las Salinas” (Alg23 and Alg24) were distanced from the main cluster in the PC1 of the PCA; however, this axis only explained 2.57% of the variation (Figure 2C). All PCs explained an equally low amount of variation, reflecting the weak population structure (Supplementary Figure 2). No clear ancestry

patterns were discernible at any K for the admixture analyses (Figure 2C, Supplementary Figure 3).

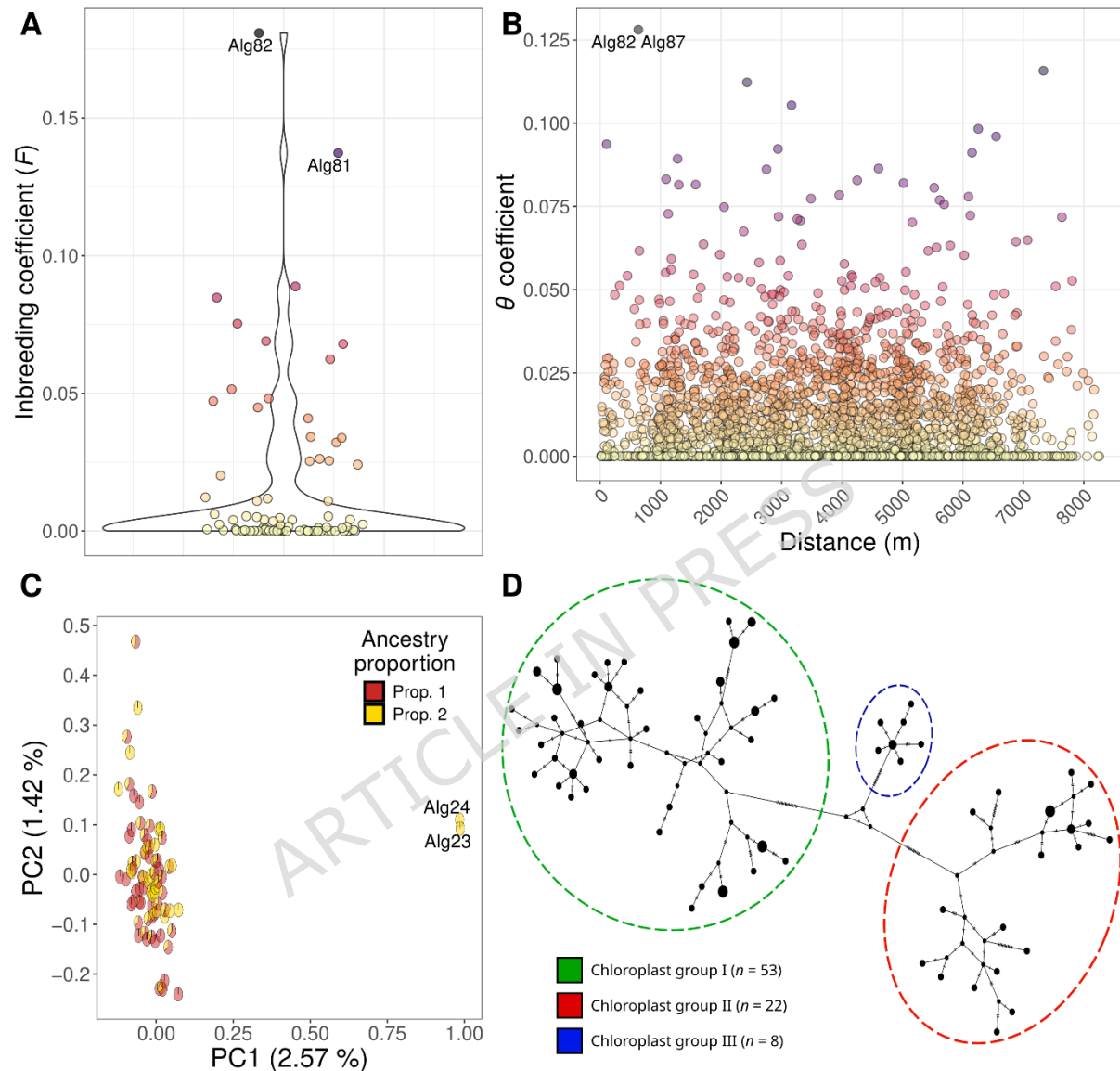


Figure 2. Inbreeding and population structure in the algarrobo population at SHBP.

A. Inbreeding coefficient (F). Highly-inbred individuals (Alg82 and Alg81) are labelled. **B.** Relatedness between pairs of individuals as measured by the kinship coefficient (θ) by their distance. A pair of second-degree relatives (Alg82 and Alg87) is labelled. **C.** Ordination of samples along the PC1 and PC2. All samples are depicted as pie charts

colored based on their ancestry proportions at $K = 2$ (model with the highest likelihood score). **D.** Median-joining chloroplast haplotype network depicting three sample groups.

Genetic diversity indices, variation, and load

Chloroplast-based network analysis resulted in three haplotype groups separated by many mutational steps without a single dominant haplotype (Figure 2D) and uncorrelated with geographical location (Supplementary Table 1). Site frequency spectra were estimated for individual samples and for the population to calculate genetic diversity indices. The average genome-wide autosomal heterozygosity was 0.19% (SD 0.024%) for the entire population (Supplementary Table S1). The per-site Watterson estimator and nucleotide diversity (π) were similar across all chromosomes, around the average 0.0026 and 0.0023, respectively (Supplementary Figure 5A, B), ranking among the lowest compared to other plant species, even those with similar reproductive strategies (Supplementary Figure 5C). Similarly, Tajima's D value averaged around -0.383 for all chromosomes in the population. Window-based genetic diversity and Tajima's D revealed several large regions with less-fluctuating low values, coinciding with a high density of repetitive elements and low density of protein-coding genes (Figure 3).

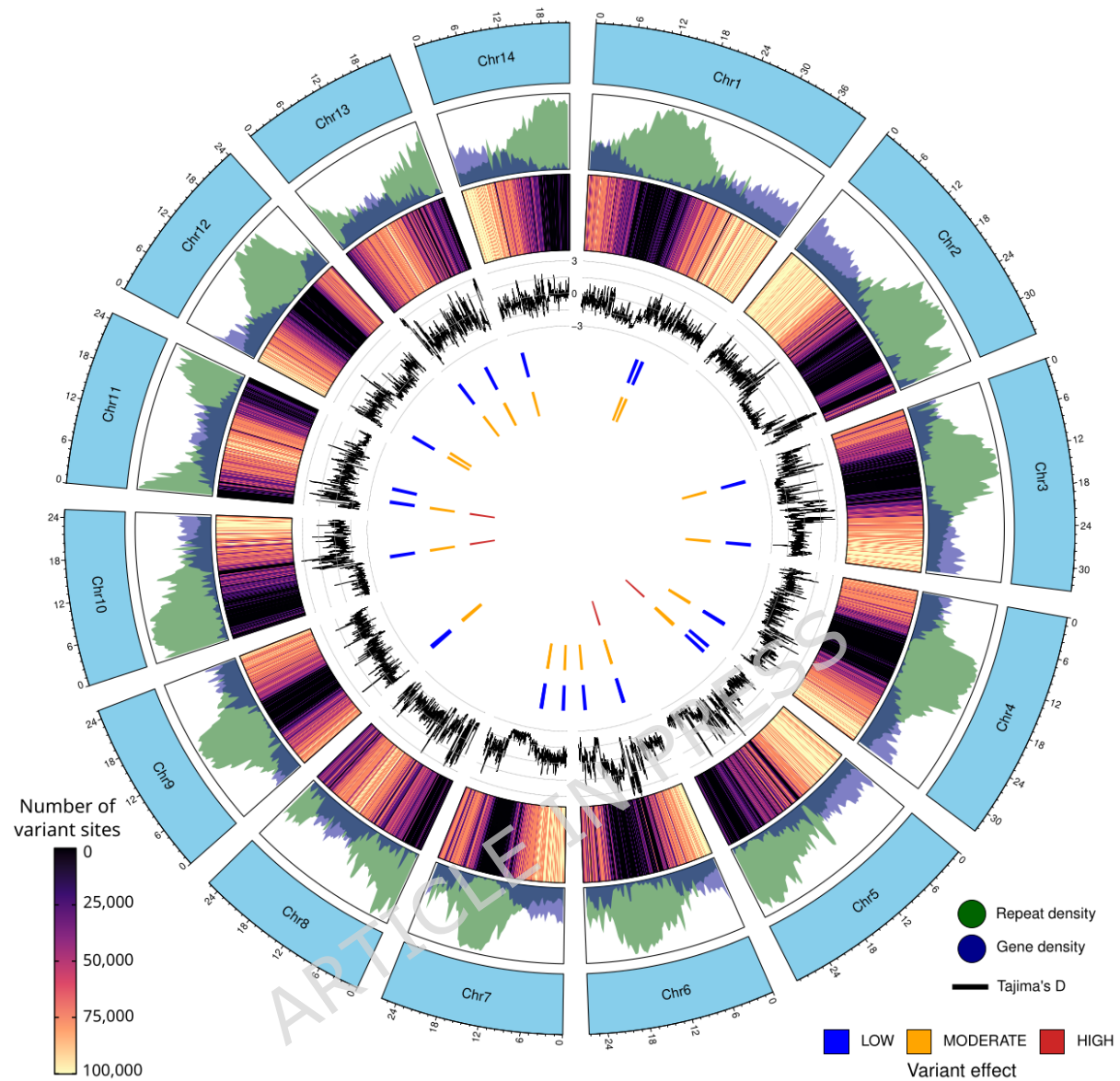


Figure 3: Genome-wide genetic variation in the algarrobo population at SHBP.

Shown from outer to inner layers are the 14 pseudochromosomes in the reference assembly, repeat and gene densities, number of variant sites in 100 kbp windows with 10 kbp step size, Tajima's D estimation in 100 kbp windows with 10 kbp step size, and the location of variants with predicted deleterious effects.

The majority of intragenic variants fell outside coding regions, thus were unlikely to directly change the protein sequence (3,225 in total, of which 476 were intronic, and 71 and 36 were located in the 3' and 5' UTR, respectively). Only a relatively small number of variants were predicted to have a deleterious effect in a set of 25 genes, with most being moderate ($n = 125$, 25 genes) and low ($n = 98$, 24 genes), and fewer having a high potential impact (4 variants in 4 genes, Table 2). Variants with potential low impact included synonymous variants ($n = 81$), premature start codon gain in the 5' UTR ($n = 9$), and substitutions within spliced regions or introns ($n = 8$; Figure 4A). All variants with potential moderate impact were missense variants ($n = 123$), with 2 being in splice regions (Figure 4A). Variants with potential high impact included stop codon gain ($n = 1$), start codon loss ($n = 1$), and two variants in the boundary of intron and exon potentially involved in splicing ($n = 2$; Figure 4A). A description of the characteristics of the four genes with a predicted high deleterious effect and genotypes for the respective variants is shown in Table 2. On average, individuals had 20.4 (SD 6.8) masked variants with low and 20.6 (SD 4.9) with moderate potential impact; and 8.4 (SD 4.2) realized variants with low and 8.3 (SD 3.1) with moderate potential impact (Figure 4B). In contrast, individuals had an average of 0.5 (SD 0.6) and 0.1 (SD 0.3) masked and realized variants with a high potential impact (Figure 4B).

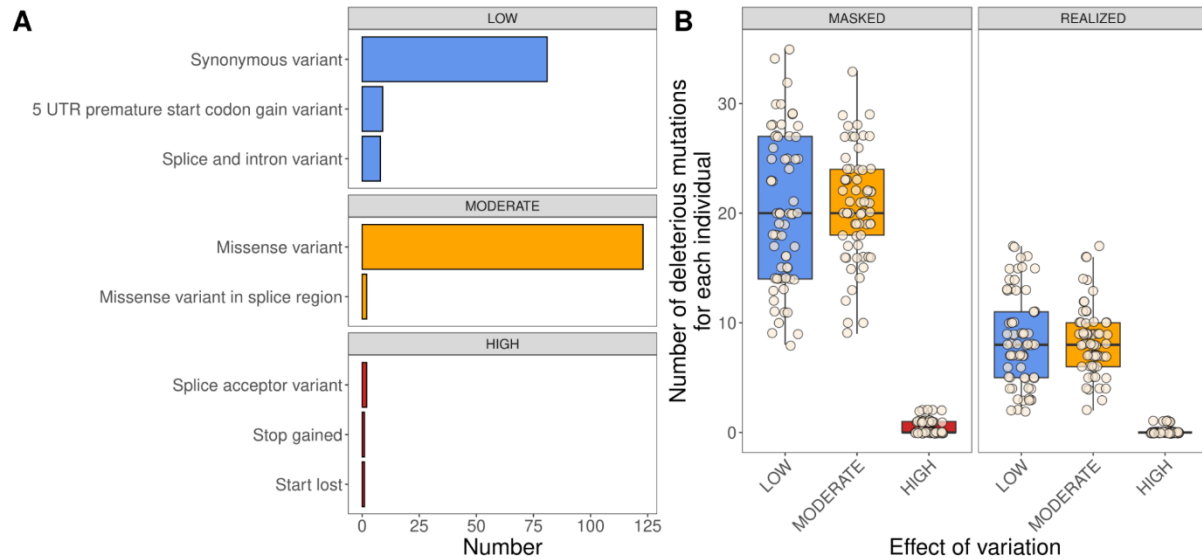


Figure 4: Genetic load in the algarrobo population at SHBP. A. Number, type and predicted deleterious impact of variants found. **B.** Amount of masked and realized variants with predicted deleterious impact across individuals.

Table 2. Description of variants with predicted high effect, genomic location and functional annotation.

Variant ^a	Effect	Genotypes ^b	Gene	Chr	Position	Annotation (EggNOG OG)	COG category*
T>G	Start loss	TT: 80 TG: 2 Unk: 1	NPAL_008776	5	3457378	Rhamnogalacturonate lyase (4JDXG)	S
G>A	Splice acceptor	GG: 73 GA: 7 Unk: 3	NPAL_010962	6	7939873	tRNA pseudouridine (4JDK7)	J
C>T	Stop gained	CC: 30 CT: 39 TT: 7 Unk: 7	NPAL_015888	10	1140347 7	UMP-CMP kinase (4JRQB)	F
G>A	Splice donor	GG: 79 AA: 1 Unk: 3	NPAL_017057	11	4009305	E3 ubiquitin-protein ligase (4JIA0)	O

^aThe allele in the right represents the allele with predicted high impact. ^bUnk: Unknown genotype. *COG category one letter code descriptions: S (Function unknown), J (Translation, ribosomal structure and biogenesis), F (Nucleotide transport and metabolism), O (Posttranslational modification, protein turnover, chaperones).

Recent demographic history

Demographic analysis using LD patterns across the population genomes revealed the highest effective population size (N_e) in the past from 200 to 175 generations ago (average N_e 256,651, SD 33,027; Figure 5A). Beginning around 175 to 150 generations ago there is a stark drop of more than 200-fold to a comparably minimal population size (average N_e 1,052, SD 217), remaining until around 110 generations ago, where the population starts rising again, reaching around 26% of the highest observed N_e within the population, and stays constant between 95 to 5 generations ago (average N_e 67,113, SD 5,394; Figure 5A). Lastly, a marked decline by 99.3% is estimated 4 generations ago, which remains constant since then (average $N_e \sim 469$, SD 53; Figure 5B).

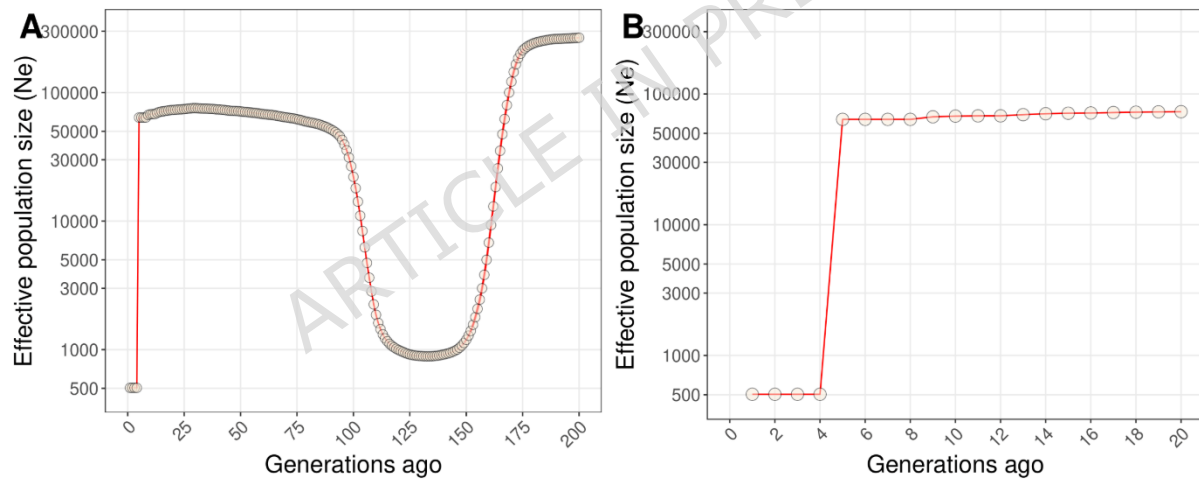


Figure 5: Historical demographic changes as inferred by GONE analysis. Average results from 100 runs and standard errors are shown for (A) 200 and (B) 20 generations ago. Results in calendar years with alternative generation times are shown in Supplementary Figure 4.

Discussion

Peruvian algarrobo populations face increasing threats yet remain poorly characterized at the molecular level, and the genomic consequences of their recent decline are still unclear. Although species-level delimitation within *Neltuma* requires further investigation using molecular markers [54], populations inhabiting the STDFs of northern Peru and extending into Ecuador are predominantly identified as *N. pallida* and are assumed to be closely related [55]. In this study, we focused on SHBP due to its high *N. pallida* density, the relatively extensive body of research conducted in the area, and its current status as a government-protected site, making it an ideal system for genomic analysis. Our results provide the first assessment of *N. pallida* genetic diversity based on genome-wide variation and dense sampling, reducing the likelihood of missing rare variants and improving the accuracy of diversity estimates [56]. Importantly, the genetic diversity metrics reported here offer a foundation for future conservation strategies at SHBP and serve as reference for studies of other, less protected and more threatened algarrobo populations in northern Peru.

Our results revealed an outbred and unstructured population (Figure 2), consistent with key aspects of Peruvian algarrobo ecology and reproductive biology. The lack of genetic structure supports the expectation that algarrobo individuals at SHBP constitute a single population, as no permanent geographical barriers limit reproduction mediated by natural pollinators, dispersers, or cattle [57-59]. Substantial human impact observed in recent decades due to deforestation and change of land use in northern Peru [11,16] may have contributed to isolate the SHBP from other algarrobo populations, thus the lack of genetic structure. In particular, the northeastern area has undergone reforestation using local specimens [60] following eviction of illegal occupants in 2009. Although larger, older trees

were preferentially sampled, some younger individuals from restored areas were also included when no nearby trees were available along the 100 m transects. Furthermore, despite the apparent genetic homogeneity of this population, future studies could leverage the large sample size and high-resolution genomic data generated here to investigate fine-scale genetic patterns and address more specific questions associated with different cohorts, successional stages, health status, and anthropogenic influence.

Low inbreeding levels are consistent with mechanisms for avoiding self-pollination documented in other *Neltuma* species. Although *Neltuma*'s anthesis was originally proposed as protogynous [61], subsequent studies have challenged this view [62,63]. Instead, a gametophytic self-incompatibility system has been proposed for *Neltuma*, supported by floral peroxidase activity and morphological traits such as binucleate pollen grains, high pollen viability, rapid germination, and the growth inhibition of incompatible pollen tubes within the style [64]. This mechanism likely evolved to maintain genetic diversity within arid zones characterized by highly fluctuating environmental conditions [10]. Indeed, many *Neltuma* populations maintain high variability despite partial selfing and limited pollen dispersal [62,65], with selection likely favoring heterozygotes suggesting an equilibrium condition [62]. Such equilibrium faces severe threats from compatible mates becoming scarce through mechanisms causing population decline and fragmentation [62]. In declining populations, flexible mating behavior may favor some degree of self-compatibility to provide reproductive assurance, buffering low-density populations against short-term extinction [66,67]. However, self-compatibility comes at the cost of inbreeding depression and heightened long-term extinction risk [68], particularly in long-lived species such as *N. pallida* where individual survival is generally

more important than reproduction. Further studies are needed to evaluate the viability and trade-offs of self-compatibility in *N. pallida* populations as an adaptation to population decline and environmental change.

Chloroplast variation supports the panmictic population structure inferred from nuclear data. Three chloroplast haplotype groups are distributed across SHBP despite the hypothesized recent bottleneck, possibly due to ongoing gene flow with external populations supplying and maintaining these lineages. Alternatively, these groups may reflect macroevolutionary processes in the history of *N. pallida*, as distinct latitudinal dispersal routes have been proposed during *Neltuma* diversification [10]. A broader analysis including multiple populations of *N. pallida* and other *Neltuma* species would help clarify the spatial and evolutionary scale of this variation.

The overall low nucleotide diversity estimation for *N. pallida* relative to other plant species (Supplementary Figure 5) coincides with low values reported for forest trees compared to other plants [69] and suggests reduced adaptive potential [70]. Given its highly variable habitat, the persistence of *N. pallida* populations may instead rely on high phenotypic plasticity [71]. For other long-lived tree species showing an adaptational lag, this could result in maladaptation [72] and delayed evolutionary responses to rapid contemporary climate change, whether adaptation or extinction [73,74]. Further surveys of genetic diversity across *N. pallida* populations, as well as analyses of metapopulation structure and gene flow, will contribute to identifying vulnerable populations and informing conservation strategies. Moreover, leveraging the broad distribution of *N. pallida* along the Peruvian coast, future studies should evaluate potential local adaptation and disentangle neutral and adaptive genetic variation. More broadly, we encourage future

studies to incorporate comparative analyses across taxa with similar life-history traits, breeding systems, and ecological niches to better contextualize patterns of genetic diversity in *N. pallida* into a broader evolutionary and ecological context.

Genome-wide diversity scans revealed regions deviating from neutral expectations, characterized by an excess or deficit of rare variants. Large regions with strongly negative Tajima's D values, such as those in chromosome 2 (10 Mbp), 7 (6 Mbp), and 10 (10 Mbp), among others, are associated with high repeat content, low gene density, and reduced number of variant sites (Figure 3). This pattern persisted even when all genomic regions were included in genotype likelihood estimation (without the region filters in Table 1), suggesting a potential centromeric function. Centromeric sequence hypervariability has been widely reported across eukaryotes [75], possibly linked to meiotic drive [76], which may influence *N. pallida* reproduction. Alternatively, identical patterns in maize have been associated with selection on genes linked to centromeres during domestication [77]. Further analyses of repetitive elements using long-read sequencing could help disentangle the roles of meiotic drive and selection shaping these signals.

Deleterious variants are typically recessive and can persist in populations as masked when heterozygous, or realized when homozygous, the latter being exposed and purged by selection [78]. Consistent with expectations under genetic load purging, only a small number of high potential impact variants were detected (Figure 4). In contrast, a moderate number of variants with predicted low and moderate deleterious impact was predicted in both masked and realized states. This pattern aligns with previous examinations of the dynamics of load in declining populations [79], where strongly deleterious variants are more easily purged, while mildly deleterious variants can increase in frequency due to

genetic drift and persist even after the population recovers. The accumulation of potential deleterious mutations may contribute to reduced plant regeneration in SHBP [80], potentially reflecting decreased fitness. The four variants with highest predicted impact were associated with changes in coding sequence length, suggesting potential disruption of protein function. Three of these variants were found at low frequencies (Table 2), consistent with purifying selection. In contrast, a stop-gain variant located in the first exon of the gene *NPAL_015888*, encoding a UMP-CMP kinase, was observed at a relatively high frequency (0.35), including seven homozygous individuals, despite producing a truncated protein of only 30 amino acids. This pattern may indicate functional redundancy, such as compensation by paralogous genes or alternative metabolic pathways involved in pyrimidine biosynthesis. Future work should evaluate how these variants affect long-term viability and their potential contribution to adaptive responses under rapid environmental change.

Generation time, the average interval between two consecutive generations in a population, is difficult to estimate precisely for long-lived tree species and is often confounded with age at maturity and maximum life span [81]. For example, using a complete life table for the tropical palm *Euterpe globosa*, an intermediate value between age at maturity (50 years) and maximum observed life span (156 years) was estimated, yielding a generation time of 101 years [81,82]. For *N. pallida*, despite occasional precocious flowering [83], reproductive maturity is typically reached between 2 and 15 years depending on environmental conditions [10], and individuals can live for more than 100 years [13,14]. Following a similar reasoning, an intermediate value of 50 years can be assumed as the generation time for *N. pallida* when interpreting our GONE results

(Figure 5). However, generation time can vary widely among forest tree species, and a more thorough analysis would be required for a precise estimate. Contemporary genome-wide low genetic diversity is consistent with the lingering effects of two historical bottlenecks inferred using GONE (Figure 5), likely reflecting the slow rate of neutral evolution in long-lived tree species [69]. Assuming a generation time of 50 years, the first major population decline (more than 200-fold) occurred around 7.5 - 8.7 kya, coinciding with significant ENSO activity during the early Holocene and extreme flood events along the Peruvian coast prior to 8 kya [84]. ENSO-related catastrophes have also been linked to the decline of pre-Columbian civilizations in northern Peru, involving flooding, anthropogenic burning and site abandonment [85,86]. Although increased precipitation due to ENSO can promote *N. pallida* growing and expansion [27,87], contemporary torrential rains also negatively affect populations through mudslides, pest outbreaks such as *E. discordis* [20], and the proliferation of suffocating vines (*Luffa* sp.), which can kill diseased and young individuals (Zaña, personal communication).

A more recent bottleneck is estimated to have occurred four generations ago (~1770 AD, assuming a generation time of 50 years). The latter Colonial to the Republican periods, from 1750 to 1950, saw a dramatic increase in demand for charcoal for fuel and timber for railways, mining, and manufacturing, leading to a massive *N. pallida* overexploitation [10]. In SHBP, deforestation has been particularly severe in northern areas and buffer zones, continuing for at least a decade after the area was formally protected in 2009 [88]. The sudden ~99.3% drop in effective population size inferred here likely reflects the delayed cumulative effects of long-term human exploitation, including pre-Columbian communities, colonial land-use change, and modern resource extraction until today

[10,11,85]. The interpretation of anthropic influence for the second bottleneck remains robust across different values of generation times (e.g., 25 and 75 years; Supplementary Figure 4). However, our N_e estimates assume that the SHBP population remained isolated for a period of time, and could be substantially biased due to population structuring [89]. Although we found no structuration at SHBP at present, gene flow with other populations and the metapopulation structure at northern Peru may have changed significantly due to human impact in recent history. Expanding sampling to include additional populations, as well as historical and ancient DNA, will help disentangle these processes and elucidate our understanding of human-driven *N. pallida* demographic change. Lastly, an artifactual drop in N_e in recent generations similar to our results is observed when populations segregate chromosomal inversions at high frequencies [89]. Given the significant role and prevalence of inversions for plant adaptation [90], employing methodological and analytical approaches to specifically target these sources of variation (e.g. [91]) will not only confirm our recent bottleneck hypothesis, but also advance our understanding of *N. pallida* genetic diversity and adaptation.

Conclusions

In this study, the genetic diversity of Peruvian algarrobos in the SHBP was surveyed to describe its current status, explore the extent of effect of recent population decline, and provide population-level genome-wide reference metrics. Inbreeding and structure analyses supported an outbred and panmictic population, consistent with *N. pallida* self-incompatibility [10] and lack of barriers between individuals in the study area. Genetic diversity was low, associated with an evident bottleneck around four generations ago, although this could represent a delayed response to older events due to *N. pallida*

longevity. Across the genome, large centromere-like regions were associated with non-neutral signals, possibly due to meiotic drive or strong selection in linked genes. Three chloroplast haplotype groups were found throughout the study area, indicating possible external inputs to the gene pool. Lastly, possible sources of adaptive potential were found both masked and realized across the genome, mostly with low and moderate effects. Altogether, these results describe the protected SHBP as a panmictic population that maintains low genetic diversity due to a recent bottleneck, which likely purged highly-deleterious recessive alleles while retaining a degree of variants with low and moderate potential impact. The extent to which these variants reflect heritable adaptability is to be determined by future studies. This information will be pivotal for *N. pallida* conservation and understanding of its unique adaptation to the extremely variable seasonally dry forests.

Declarations

Ethics approval and consent to participate. Not applicable

Consent for publication. Not applicable.

Availability of data and materials. All data generated or analyzed during this study are available in NCBI under BioProject PRJNA1452894.

Supplementary Figure 1. Relatedness between pairs of individuals as measured by the KING coefficient by their distance. A pair of second-degree relatives (Alg82 and Alg87), as measured by the kinship coefficient (θ), is labelled.

Supplementary Figure 2. A. Distribution of variance percentage across all PCs. **B.** Ordination of samples along the PC2 and PC3. All samples are depicted as pie charts colored based on their ancestry proportions ($K = 2$).

Supplementary Figure 3. Population structure obtained from NGSAdmix. Individual ancestry proportions for different numbers of genetic clusters are grouped according to the SHBP sectors as described by the Servicio Nacional de Áreas Naturales Protegidas por el Estado (SERNANP) [30,92]. The run with the highest likelihood was used for each K displayed (likelihood score shown in parenthesis).

Supplementary Figure 4. Demographic inference using GONE analysis for alternative generation time values. Results are shown as averages and standard errors from 100 runs for 200 generations assuming generation times of (A) 25, (B) 50, and (C) 75 years, and for 20 generations assuming generation times of (D) 25, (E) 50, and (F) 75 years. Discontinuous black lines delimit historical eras from left to right: Republican (0 to 201 years ago), Colonial (201 to 490 years ago) and Pre-Hispanic (490 to 15000 years ago). Historical periods from past to present within the Pre-Hispanic era correspond to the Lithic (9020 to 15020 years ago), Archaic (3820 to 9020 years ago), Early Horizon (2220 to 3820 years ago), Early Intermediate (1420 to 2220 years ago), Middle Horizon (1020 to 2220 years ago), Late Intermediate (584 to 1020 years ago), and Late Horizon or Inca Empire (490 to 584 years ago). Within the Colonial era, historical periods are the Conquest Process (480 to 490 years ago) and Viceroyalty of Peru (201 to 480 years ago). Lastly, historical periods within the Republican era correspond to the Independence Process (198 to 201 years ago), Early Industrialization (143 to 177 years ago), War of the Pacific (139 to 143 years ago), National Reconstruction (103 to 138 years ago), Leguía's

Presidency (92 to 103 years ago), and Contemporary Republican Period (0 to 92 years ago).

Supplementary Figure 5. Average per-site nucleotide diversity estimates for *N. pallida* at SHBP as measured by **A.** π and **B.** θ_w coefficients. **C.** Average nucleotide diversity for different diploid terrestrial plants by their mating system. Data was obtained from Leffler et al. (2012) [93]. Chromosome numbers were obtained from ploidDB [94] and are shown in parenthesis. The number of populations with available data is indicated above each data point. For species with more than one population, the mean and error bars are shown.

Supplementary Table 1. Overview of samples used in this study. The Status column describes the health status as judged by the field collectors. Sector refers to the location of the tree according to the sectorization by SERNANP [30]. Coordinates are shown in the Universal Transverse Mercator (UTM) system and World Geodetic System (WGS) 84. DNA isolations results are shown as concentration in (ng/ μ l) and purity (ratio of absorbances at 260 nm and 280 nm, and 260 nm and 230 nm). Sequencing results are shown as the raw number of nucleotides sequenced and the achieved depth and breadth of coverage, both at the nuclear and chloroplast level. Individuals removed are indicated in the “Removed?” column. Inbreeding coefficient (F) and individual autosomal heterozygosity (H) are shown in columns F and H, respectively.

Supplementary Table 2. Detailed sequencing and mapping results for the nuclear and chloroplast reference genomes based on the summary report from Paleomix v1.3.7 [35]. Individuals highlighted in red were removed due to a low depth. Individuals highlighted in

yellow were closely-related as measured by the kinship coefficient (θ), from which Alg82 was removed.

Supplementary Table 3. GONE results across 100 runs. For each run, the geometric mean of 40 internal replicates is shown, using up to 50,000 SNPs per chromosome with $MAF \geq 0.01$.

Competing interests. The authors declare that they have no competing interests.

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Author's contributions. AC and OG designed the study and provided the resources. LR, CE, OG, and SM collected the samples. LR, CE, and SC, processed the samples in the laboratory. LR and BV performed the bioinformatic analyses. LR wrote the manuscript draft. All authors read and approved the final manuscript.

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